

Final Exam

Macroeconomics II

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Problem 1 (Questions 1–20): Expectations, NKPC, shocks

① Definition. If $p_t \equiv \log P_t$, then inflation is A

- (A) $\pi_t \equiv p_t - p_{t-1}$
- (B) $\pi_t \equiv P_t - P_{t-1}$
- (C) $\pi_t \equiv \log P_t$
- (D) $\pi_t \equiv p_{t+1} - p_t$

Explanations.

- (A) Correct.
- (B) Wrong: depends on units; not the model definition.
- (C) Wrong: that is the log price level.
- (D) Wrong timing.

② Three inflation objects. Which pair must be distinguished in NK models? B

- (A) π_t and i_t
- (B) π_t and $\mathbb{E}_t \pi_{t+1}$
- (C) x_t and u_t
- (D) β and κ

Explanations.

- (A) Not the key expectations distinction.
- (B) Correct.
- (C) Different objects, but not the core expectations issue.
- (D) Parameters.

③ Adaptive expectations. Under adaptive expectations, we set B

- (A) $\pi_t^e = \mathbb{E}_t \pi_{t+1}$
- (B) $\pi_t^e = \pi_{t-1}$
- (C) $\pi_t^e = \pi^*$ always
- (D) $\pi_t^e = 0$ always

Explanations.

- (A) Forward-looking closure.
- (B) Correct.
- (C) Anchored benchmark.
- (D) Not standard.

④ Forward-looking closure. A convenient NK choice of is B

- (A) $\pi_t^e = \pi_{t-1}$
- (B) $\pi_t^e = \mathbb{E}_t \pi_{t+1}$
- (C) $\pi_t^e = \pi_t$
- (D) $\pi_t^e = i_t$

Explanations.

- (A) Adaptive.
- (B) Correct.
- (C) Not an expectation.
- (D) Wrong.

⑤ NKPC. The New Keynesian Phillips Curve is A

- (A) $\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t$
- (B) $\pi_t = \pi_{t-1} + \kappa x_t + u_t$
- (C) $\pi_t = \kappa x_t + u_t$
- (D) $\pi_t = \beta \pi_{t-1} - \kappa x_t$

Explanations.

- (A) Correct.
- (B) Backward-looking PC.
- (C) Missing expectations.
- (D) Wrong sign/objects.

⑥ Expectations matter even if slack is zero. If $x_t = 0$ and $u_t = 0$, NKPC implies B

- (A) $\pi_t = 0$ always
- (B) $\pi_t = \beta \mathbb{E}_t \pi_{t+1}$
- (C) $\pi_t = \pi_{t-1}$
- (D) $\pi_t = \kappa$

Explanations.

- (A) Wrong unless expected inflation is zero.
- (B) Correct.
- (C) Adaptive closure.
- (D) Wrong.

⑦ Meaning of x_t . In this course, the output gap is B

- (A) $x_t \equiv y_t + y_t^{FE}$
- (B) $x_t \equiv y_t - y_t^{FE}$
- (C) $x_t \equiv \pi_t - \pi^*$
- (D) $x_t \equiv i_t - \pi_t$

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Inflation gap.
- (D) Real-rate object.

⑧ Interpretation of β in NKPC. A larger β means A

- (A) more forward-looking inflation (greater weight on expected future inflation)
- (B) steeper NKPC (larger response to x_t)
- (C) tougher monetary policy
- (D) larger cost-push shocks

Explanations.

- (A) Correct.
- (B) That is κ .
- (C) That is ϕ_π .
- (D) That is u_t .

⑨ Interpretation of κ in NKPC. A larger κ means B

- (A) more weight on expected inflation
- (B) inflation responds more to slack x_t
- (C) the central bank reacts more to inflation
- (D) the natural real rate rises

Explanations.

- (A) That is β .
- (B) Correct.
- (C) That is ϕ_π .
- (D) Not implied.

⑩ Cost-push shock. In NKPC, u_t is best interpreted as B

- (A) a demand shock moving the IS curve
- (B) a markup/energy/bottleneck shock shifting inflation for given x_t
- (C) the inflation target
- (D) the nominal interest rate

Explanations.

- (A) Wrong block.
- (B) Correct.
- (C) Target is π^* .
- (D) Wrong.

⑪ Calvo pricing. Each period, a fraction of firms that can reset prices is B

- (A) θ
- (B) $1 - \theta$
- (C) β
- (D) κ

Explanations.

- (A) Wrong: θ keep price fixed.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

⑫ Calvo intuition. A firm that resets at time t chooses $P_t^\#$ to maximize profits while A

- (A) the price remains fixed with some probability
- (B) money supply is constant
- (C) inflation is exactly zero
- (D) the output gap is zero

Explanations.

- (A) Correct.
- (B) Not the condition.
- (C) Not the objective.
- (D) Not required.

⑬ Shift in NKPC. Holding x_t fixed, an increase in $\mathbb{E}_t \pi_{t+1}$ B

- (A) lowers π_t one-for-one
- (B) raises π_t by β times the change
- (C) does not affect π_t
- (D) changes κ

Explanations.

- (A) Wrong sign.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

⑭ Demand shock comovement. A positive demand shock (raising x_t) tends to generate A

- (A) $x_t \uparrow$ and $\pi_t \uparrow$
- (B) $x_t \downarrow$ and $\pi_t \uparrow$
- (C) $x_t \uparrow$ and $\pi_t \downarrow$
- (D) $x_t \downarrow$ and $\pi_t \downarrow$

Explanations.

- (A) Correct.
- (B) Cost-push pattern.
- (C) Wrong.
- (D) Contractionary demand.

⑮ Cost-push comovement. A positive cost-push shock $u_t > 0$ (with stabilization) tends to generate B

- (A) $x_t \uparrow$ and $\pi_t \uparrow$
- (B) $x_t \downarrow$ and $\pi_t \uparrow$
- (C) $x_t \uparrow$ and $\pi_t \downarrow$
- (D) $x_t \downarrow$ and $\pi_t \downarrow$

Explanations.

- (A) Not typical.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

⑩ Calculation (NKPC). Let $\beta = 0.98$, $\mathbb{E}_t \pi_{t+1} = 3\%$, $\kappa = 0.5$, $x_t = -2\%$, and $u_t = 0$. Then π_t is A

- (A) 1.94%
- (B) 2.94%
- (C) 3.94%
- (D) 0.50%

Explanations.

- (A) Correct: $0.98 \cdot 3 - 1 = 1.94$.
- (B) Wrong: ignores x_t .
- (C) Wrong sign.
- (D) Wrong.

⑪ Calculation (solve for slack). Suppose $\pi_t = 3\%$, $\beta = 1$, $\mathbb{E}_t \pi_{t+1} = 2\%$, $\kappa = 0.5$, and $u_t = 0$. Then x_t equals A

- (A) 2%
- (B) 1%
- (C) 0%
- (D) -1%

Explanations.

- (A) Correct: $3 - 2 = 0.5x \Rightarrow x = 2$.
- (B) Wrong.
- (C) Wrong.

(D) Wrong.

- ⑮ Stabilizing a one-time cost-push shock. If $\mathbb{E}_t \pi_{t+1} = \pi^*$ and the central bank wants $\pi_t = \pi^*$ with $u_t = 1\%$ and $\kappa = 0.25$, then it must set A

(A) $x_t = -4\%$

(B) $x_t = -1\%$

(C) $x_t = +4\%$

(D) $x_t = 0\%$

Explanations.

(A) Correct: $x = -u/\kappa = -4$.

(B) Wrong magnitude.

(C) Wrong sign.

(D) Wrong.

- ⑯ Anchored vs drifting. Expectations are drifting if B

(A) after a one-time shock, $\mathbb{E}_t \pi_{t+1}$ returns to π^*

(B) after a one-time shock, $\mathbb{E}_t \pi_{t+1}$ stays above π^*

(C) $x_t = 0$ always

(D) $u_t = 0$ always

Explanations.

(A) Anchored.

(B) Correct.

(C) Not definition.

(D) Not definition.

- ⑰ Key contrast. Relative to adaptive expectations, forward-looking expectations make inflation more sensitive to B

- (A) last period inflation mechanically
- (B) beliefs about future policy (credibility) today
- (C) money demand only
- (D) the money supply only

Explanations.

- (A) Adaptive feature.
 - (B) Correct.
 - (C) Wrong.
 - (D) Wrong.
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Problem 2 (Questions 21–40): NK IS, Taylor rule, determinacy, ZLB

- (21) Fisher (ex-ante real rate). The expected real policy rate is approximately B
- (A) $i_t + \mathbb{E}_t \pi_{t+1}$
 - (B) $i_t - \mathbb{E}_t \pi_{t+1}$
 - (C) $\pi_t - i_t$
 - (D) $i_t - \pi_t$

Explanations.

- (A) Wrong sign.
- (B) Correct.
- (C) Wrong sign.
- (D) Uses current inflation.

- (22) NK IS (Euler) equation. A standard gap-form NK IS is A

- (A) $x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\sigma} (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n)$
- (B) $x_t = \mathbb{E}_t x_{t+1} + \frac{1}{\sigma} (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n)$
- (C) $x_t = -\sigma (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n)$
- (D) $x_t = \kappa (i_t - \mathbb{E}_t \pi_{t+1})$

Explanations.

- (A) Correct.
- (B) Wrong sign.
- (C) Wrong scaling.
- (D) Wrong.

②③ Interpretation of σ . A larger σ implies the output gap responds B

- (A) more strongly to the real-rate gap
- (B) less strongly to the real-rate gap
- (C) only to u_t
- (D) only to κ

Explanations.

- (A) Slope is $1/\sigma$.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

②④ Natural real rate. r_t^n is best interpreted as B

- (A) the nominal policy rate
- (B) the flexible-price real rate that would make $x_t = 0$
- (C) the inflation target
- (D) the cost-push shock

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

(25) Calculation (NK IS). Suppose $\mathbb{E}_t x_{t+1} = 0$, $\sigma = 2$, $i_t = 4\%$, $\mathbb{E}_t \pi_{t+1} = 2\%$, and $r_t^n = 1\%$. Then x_t is A

- (A) -0.5%
- (B) -1%
- (C) $+1\%$
- (D) 0%

Explanations.

- (A) Correct.
- (B) Forgot $1/\sigma$.
- (C) Wrong sign.
- (D) Wrong.

(26) Calculation (real rate). If $i_t = 3\%$ and $\mathbb{E}_t \pi_{t+1} = 1.5\%$, then the expected real rate is B

- (A) 4.5%
- (B) 1.5%
- (C) -1.5%
- (D) 3%

Explanations.

- (A) Adds instead of subtracts.
- (B) Correct.
- (C) Wrong sign.
- (D) Nominal.

(27) Forward guidance channel. Holding i_t fixed, a higher $\mathbb{E}_t \pi_{t+1}$ implies B

- (A) higher real rate and lower x_t
- (B) lower real rate and higher x_t
- (C) no change in x_t
- (D) x_t must fall because expectations are higher

Explanations.

- (A) Wrong sign.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

(28) Taylor rule (general form). A standard rule in the NK core is A

- (A) $i_t = \bar{i} + \phi_\pi(\pi_t - \pi^*) + \phi_x x_t$
- (B) $i_t = \pi_t - \phi_\pi x_t$
- (C) $i_t = r_t^n$ always
- (D) $i_t = \mathbb{E}_t \pi_{t+1}$

Explanations.

- (A) Correct.
- (B) Wrong.
- (C) Wrong.
- (D) Wrong.

(29) Taylor principle in NK. A standard sufficient condition for determinacy is B

- (A) $\phi_\pi < 1$
- (B) $\phi_\pi > 1$
- (C) $\phi_x < 0$
- (D) $\kappa = 0$

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Not the standard condition.
- (D) Not sufficient.

30 Policy experiment sequence. After a one-time cost-push shock $u_t > 0$, the NK logic is A

- (A) NKPC pushes $\pi_t \uparrow$; rule raises i_t ; NK IS yields $x_t \downarrow$; NKPC then helps bring π_t down
- (B) NKPC pushes $\pi_t \uparrow$; rule lowers i_t ; $x_t \uparrow$; inflation falls immediately
- (C) rule does nothing; inflation returns automatically
- (D) NK IS implies $x_t \uparrow$ when i_t rises

Explanations.

- (A) Correct.
- (B) Wrong.
- (C) Wrong.
- (D) Wrong.

31 ZLB constraint is B

- (A) $r_t \geq 0$
- (B) $i_t \geq 0$
- (C) $\pi_t \geq 0$
- (D) $x_t \geq 0$

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Wrong.

(D) Wrong.

32) ZLB calculation. A rule implies $i_t = -0.5\%$, but ZLB binds so $i_t = 0$. If $\mathbb{E}_t \pi_{t+1} = 1\%$, then the expected real rate is B

(A) -1.5%

(B) -1%

(C) 0%

(D) $+1\%$

Explanations.

(A) Wrong: use $i = 0$.

(B) Correct: $0 - 1 = -1$.

(C) Wrong.

(D) Wrong sign.

33) ZLB amplification. At the ZLB, a fall in expected inflation tends to B

(A) lower the real rate and raise demand

(B) raise the real rate and lower demand

(C) leave the real rate unchanged

(D) eliminate the NKPC

Explanations.

(A) Wrong sign.

(B) Correct.

(C) Wrong.

(D) Wrong.

34) Three-equation NK core consists of A

(A) IS (Euler), NKPC, Taylor rule

(B) IS, LM, SRAS

- (C) AD, SRAS, LRAS
- (D) budget constraint, money demand, IS

Explanations.

- (A) Correct.
- (B) Wrong.
- (C) Wrong.
- (D) Wrong.

- ③⑤ **Sacrifice ratio (simple).** If expectations are anchored ($\mathbb{E}_t \pi_{t+1} = \pi^*$) and $u_t = 0$, then reducing inflation by 1 pp requires A

- (A) $x_t = -1/\kappa$ pp
- (B) $x_t = -\kappa$ pp
- (C) $x_t = +1/\kappa$ pp
- (D) $x_t = 0$

Explanations.

- (A) Correct.
- (B) Wrong.
- (C) Wrong sign.
- (D) Wrong.

- ③⑥ **Sacrifice ratio (numeric).** If $\kappa = 0.2$ and you want inflation 0.5 pp lower (anchored expectations, $u_t = 0$), required x_t is A

- (A) -2.5%
- (B) -0.1%
- (C) $+2.5\%$
- (D) -10%

Explanations.

- (A) Correct.

- (B) Wrong.
- (C) Wrong sign.
- (D) Wrong.

37) 2-period equilibrium (calc). Suppose $\mathbb{E}_0\pi_1 = \pi^* = 2\%$, $\mathbb{E}_0x_1 = 0$, $\beta = 1$, $\kappa = 0.5$, $u_0 = 1\%$. Policy: $i_0 = r_0^n + \mathbb{E}_0\pi_1 + \phi_\pi(\pi_0 - \pi^*) + \phi_x x_0$ with $\sigma = 1$, $\phi_\pi = 2$, $\phi_x = 1$. Then π_0 is closest to C

- (A) 2.0%
- (B) 2.33%
- (C) 2.67%
- (D) 3.0%

Explanations.

- (A) Wrong.
- (B) Wrong.
- (C) Correct: $\pi_0 \approx 2.67\%$.
- (D) Too high.

38) In the same 2-period equilibrium as previous question, x_0 is closest to C

- (A) 0.0%
- (B) -0.33%
- (C) -0.67%
- (D) -2.0%

Explanations.

- (A) Wrong.
- (B) Wrong.
- (C) Correct: $x_0 \approx -0.67\%$.
- (D) Too large.

39) Determinacy intuition. The Taylor principle helps anchor inflation because it ensures that if π_t rises, policy B

- (A) lowers the real rate, validating higher inflation
- (B) raises the real rate enough to create $x_t < 0$, pushing inflation down via NKPC
- (C) keeps the real rate fixed at r_t^n regardless of inflation
- (D) eliminates cost-push shocks

Explanations.

- (A) Destabilizing.
- (B) Correct.
- (C) That is $\phi_\pi = 1$; not anchoring.
- (D) Policy cannot delete shocks.

40) Old vs new Phillips curve. Relative to the accelerationist PC, the NKPC replaces A

- (A) π_{t-1} with $\mathbb{E}_t \pi_{t+1}$
- (B) x_t with y_t
- (C) u_t with i_t
- (D) κ with β

Explanations.

- (A) Correct.
- (B) Wrong.
- (C) Wrong.
- (D) Wrong.